

1. Introduction to Email Spam Detection

Email spam detection is the process of identifying unwanted, suspicious, misleading, or potentially harmful email messages. A spam filter examines different signals such as the subject line, message text, links, sender information, attachments, and email headers.

Keywords can be useful signals, but a single word should not automatically classify an email as spam. Modern filters combine many signals to reduce false positives.

Important Keywords / Signals

No.	Keyword / Phrase	Why it may be suspicious
1	spam	Attention-grabbing or promotional language.
2	promotion	May create a reward-based incentive.
3	offer	Can create pressure to respond quickly.
4	winner	May request action, payment, or information.
5	urgent	Could be used in misleading offers or scams.
6	free	Attention-grabbing or promotional language.
7	prize	May create a reward-based incentive.
8	claim	Can create pressure to respond quickly.
9	bonus	May request action, payment, or information.

Note: A keyword is only a signal. Legitimate emails can contain these words too, so detection should consider context and multiple indicators.

2. Common Promotional Spam Keywords

Promotional spam often tries to attract attention by promising discounts, gifts, or special deals. These words are not always harmful because legitimate businesses also use them. The surrounding context matters.

Important Keywords / Signals

No.	Keyword / Phrase	Why it may be suspicious
1	FREE	Attention-grabbing or promotional language.
2	LIMITED OFFER	May create a reward-based incentive.
3	SPECIAL OFFER	Can create pressure to respond quickly.
4	BIG DISCOUNT	May request action, payment, or information.
5	SALE	Could be used in misleading offers or scams.
6	DEAL	Attention-grabbing or promotional language.
7	BONUS	May create a reward-based incentive.
8	SAVE NOW	Can create pressure to respond quickly.
9	EXCLUSIVE	May request action, payment, or information.
10	ACT NOW	Could be used in misleading offers or scams.

Note: A keyword is only a signal. Legitimate emails can contain these words too, so detection should consider context and multiple indicators.

3. Prize and Lottery Keywords

Prize and lottery scams commonly claim that the recipient has won something despite not entering a competition. Such messages may request personal information, payment, or a click on a suspicious link.

Important Keywords / Signals

No.	Keyword / Phrase	Why it may be suspicious
1	YOU WON	Attention-grabbing or promotional language.
2	WINNER	May create a reward-based incentive.
3	CONGRATULATIONS	Can create pressure to respond quickly.
4	LOTTERY	May request action, payment, or information.
5	JACKPOT	Could be used in misleading offers or scams.
6	PRIZE	Attention-grabbing or promotional language.
7	REWARD	May create a reward-based incentive.
8	LUCKY WINNER	Can create pressure to respond quickly.
9	CLAIM YOUR PRIZE	May request action, payment, or information.
10	SELECTED	Could be used in misleading offers or scams.

Note: A keyword is only a signal. Legitimate emails can contain these words too, so detection should consider context and multiple indicators.

4. Urgency and Pressure Keywords

Scammers often create urgency to make recipients act before thinking carefully. Urgency alone is not proof of spam, because legitimate security alerts can also be urgent.

Important Keywords / Signals

No.	Keyword / Phrase	Why it may be suspicious
1	URGENT	Attention-grabbing or promotional language.
2	IMMEDIATE ACTION	May create a reward-based incentive.
3	ACT NOW	Can create pressure to respond quickly.
4	LAST CHANCE	May request action, payment, or information.
5	FINAL NOTICE	Could be used in misleading offers or scams.
6	RESPOND NOW	Attention-grabbing or promotional language.
7	TIME SENSITIVE	May create a reward-based incentive.
8	DO NOT DELAY	Can create pressure to respond quickly.
9	EXPIRING	May request action, payment, or information.
10	TODAY ONLY	Could be used in misleading offers or scams.

Note: A keyword is only a signal. Legitimate emails can contain these words too, so detection should consider context and multiple indicators.

5. Money and Financial Keywords

Financial spam may involve fake refunds, investments, loans, payments, or requests for money. These messages deserve extra attention when they ask for credentials or financial information.

Important Keywords / Signals

No.	Keyword / Phrase	Why it may be suspicious
1	CASH	Attention-grabbing or promotional language.
2	MONEY	May create a reward-based incentive.
3	PAYMENT	Can create pressure to respond quickly.
4	REFUND	May request action, payment, or information.
5	LOAN	Could be used in misleading offers or scams.
6	INVESTMENT	Attention-grabbing or promotional language.
7	PROFIT	May create a reward-based incentive.
8	EARN MONEY	Can create pressure to respond quickly.
9	BANK TRANSFER	May request action, payment, or information.
10	WIRE TRANSFER	Could be used in misleading offers or scams.

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6. Banking and Account Keywords

Phishing emails frequently imitate banks, payment services, or account-management systems. They may ask users to verify information or sign in through a link.

Important Keywords / Signals

No.	Keyword / Phrase	Why it may be suspicious
1	VERIFY ACCOUNT	Attention-grabbing or promotional language.
2	ACCOUNT SUSPENDED	May create a reward-based incentive.
3	BANK ALERT	Can create pressure to respond quickly.
4	SECURITY ALERT	May request action, payment, or information.
5	CONFIRM IDENTITY	Could be used in misleading offers or scams.
6	UPDATE ACCOUNT	Attention-grabbing or promotional language.
7	LOGIN REQUIRED	May create a reward-based incentive.
8	UNUSUAL ACTIVITY	Can create pressure to respond quickly.
9	PAYMENT FAILED	May request action, payment, or information.

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7. Password and Credential Keywords

Credential phishing attempts often target passwords, usernames, one-time codes, and account recovery information. A legitimate service should be checked through its official website or app rather than through an unexpected email link.

Important Keywords / Signals

No.	Keyword / Phrase	Why it may be suspicious
1	PASSWORD	Attention-grabbing or promotional language.
2	USERNAME	May create a reward-based incentive.
3	LOGIN	Can create pressure to respond quickly.
4	SIGN IN	May request action, payment, or information.
5	VERIFY PASSWORD	Could be used in misleading offers or scams.
6	SECURITY CODE	Attention-grabbing or promotional language.
7	OTP	May create a reward-based incentive.
8	ONE-TIME CODE	Can create pressure to respond quickly.
9	ACCOUNT VERIFICATION	May request action, payment, or information.

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8. Personal Information Keywords

Some suspicious emails request sensitive personal information. These requests should be treated carefully, especially when the email is unexpected or asks for information that the sender should already have.

Important Keywords / Signals

No.	Keyword / Phrase	Why it may be suspicious
1	PERSONAL INFORMATION	Attention-grabbing or promotional language.
2	DATE OF BIRTH	May create a reward-based incentive.
3	ID NUMBER	Can create pressure to respond quickly.
4	IDENTITY	May request action, payment, or information.
5	ADDRESS	Could be used in misleading offers or scams.
6	PHONE NUMBER	Attention-grabbing or promotional language.
7	SOCIAL SECURITY	May create a reward-based incentive.
8	TAX ID	Can create pressure to respond quickly.
9	DOCUMENT REQUIRED	May request action, payment, or information.

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9. Gift and Freebie Keywords

Free-gift messages are common in spam because they use curiosity and reward-based language. A message offering a free item may be legitimate, but unexpected offers should be verified independently.

Important Keywords / Signals

No.	Keyword / Phrase	Why it may be suspicious
1	FREE GIFT	Attention-grabbing or promotional language.
2	FREE COUPON	May create a reward-based incentive.
3	FREE SAMPLE	Can create pressure to respond quickly.
4	GIFT CARD	May request action, payment, or information.
5	VOUCHER	Could be used in misleading offers or scams.
6	FREE TICKET	Attention-grabbing or promotional language.
7	FREE TRIAL	May create a reward-based incentive.
8	CLAIM FREE	Can create pressure to respond quickly.
9	BONUS GIFT	May request action, payment, or information.

Note: A keyword is only a signal. Legitimate emails can contain these words too, so detection should consider context and multiple indicators.

10. Investment and Crypto Spam Keywords

Investment spam may promise unusually high returns with little or no risk. Cryptocurrency scams can use similar language. Keywords are only one indicator; financial claims should be independently verified.

Important Keywords / Signals

No.	Keyword / Phrase	Why it may be suspicious
1	GUARANTEED PROFIT	Attention-grabbing or promotional language.
2	HIGH RETURN	May create a reward-based incentive.
3	RISK FREE	Can create pressure to respond quickly.
4	CRYPTO	May request action, payment, or information.
5	BITCOIN	Could be used in misleading offers or scams.
6	FOREX	Attention-grabbing or promotional language.
7	TRADING SIGNAL	May create a reward-based incentive.
8	DOUBLE YOUR MONEY	Can create pressure to respond quickly.
9	PASSIVE INCOME	May request action, payment, or information.

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11. Job and Work-from-Home Spam

Fake job emails may promise easy work, unusually high salaries, or quick hiring. Warning signs include requests for fees, banking information, or identity documents before a legitimate hiring process.

Important Keywords / Signals

No.	Keyword / Phrase	Why it may be suspicious
1	WORK FROM HOME	Attention-grabbing or promotional language.
2	EASY MONEY	May create a reward-based incentive.
3	NO EXPERIENCE	Can create pressure to respond quickly.
4	HIRING NOW	May request action, payment, or information.
5	HIGH SALARY	Could be used in misleading offers or scams.
6	QUICK HIRING	Attention-grabbing or promotional language.
7	PART TIME	May create a reward-based incentive.
8	MAKE MONEY ONLINE	Can create pressure to respond quickly.
9	JOB OFFER	May request action, payment, or information.

Note: A keyword is only a signal. Legitimate emails can contain these words too, so detection should consider context and multiple indicators.

12. Health and Miracle-Claim Spam

Spam messages may promote unverified health products or make unrealistic claims. Words suggesting guaranteed results should be treated as warning signals rather than proof.

Important Keywords / Signals

No.	Keyword / Phrase	Why it may be suspicious
1	MIRACLE CURE	Attention-grabbing or promotional language.
2	GUARANTEED RESULTS	May create a reward-based incentive.
3	LOSE WEIGHT FAST	Can create pressure to respond quickly.
4	INSTANT CURE	May request action, payment, or information.
5	100% EFFECTIVE	Could be used in misleading offers or scams.
6	DOCTOR RECOMMENDED	Attention-grabbing or promotional language.
7	SECRET FORMULA	May create a reward-based incentive.
8	NO SIDE EFFECTS	Can create pressure to respond quickly.

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13. Romance and Social-Engineering Spam

Social engineering uses emotions such as trust, fear, curiosity, or sympathy to influence recipients. Suspicious messages may quickly create an emotional connection or ask for money or personal information.

Important Keywords / Signals

No.	Keyword / Phrase	Why it may be suspicious
1	DEAR FRIEND	Attention-grabbing or promotional language.
2	I NEED YOUR HELP	May create a reward-based incentive.
3	TRUST ME	Can create pressure to respond quickly.
4	PRIVATE	May request action, payment, or information.
5	CONFIDENTIAL	Could be used in misleading offers or scams.
6	PLEASE HELP	Attention-grabbing or promotional language.
7	SEND MONEY	May create a reward-based incentive.
8	EMERGENCY	Can create pressure to respond quickly.
9	KEEP THIS SECRET	May request action, payment, or information.

Note: A keyword is only a signal. Legitimate emails can contain these words too, so detection should consider context and multiple indicators.

14. Suspicious Link and Attachment Signals

Links and attachments can be dangerous when they are unexpected. Keywords may indicate a request to open a document, download software, or follow a link. The actual URL and attachment type should also be inspected.

Important Keywords / Signals

No.	Keyword / Phrase	Why it may be suspicious
1	CLICK HERE	Attention-grabbing or promotional language.
2	OPEN ATTACHMENT	May create a reward-based incentive.
3	DOWNLOAD NOW	Can create pressure to respond quickly.
4	VIEW DOCUMENT	May request action, payment, or information.
5	VERIFY HERE	Could be used in misleading offers or scams.
6	ACCESS FILE	Attention-grabbing or promotional language.
7	ENABLE MACROS	May create a reward-based incentive.
8	DOWNLOAD FILE	Can create pressure to respond quickly.
9	CLICK TO CLAIM	May request action, payment, or information.

Note: A keyword is only a signal. Legitimate emails can contain these words too, so detection should consider context and multiple indicators.

15. Fake Invoice and Payment Spam

Fake invoice emails can imitate companies or suppliers and request payment. A recipient should verify invoices using known contact details rather than replying to the suspicious email.

Important Keywords / Signals

No.	Keyword / Phrase	Why it may be suspicious
1	INVOICE ATTACHED	Attention-grabbing or promotional language.
2	PAYMENT DUE	May create a reward-based incentive.
3	OVERDUE PAYMENT	Can create pressure to respond quickly.
4	BILLING NOTICE	May request action, payment, or information.
5	PURCHASE ORDER	Could be used in misleading offers or scams.
6	PAY NOW	Attention-grabbing or promotional language.
7	PAYMENT REQUIRED	May create a reward-based incentive.
8	UNPAID INVOICE	Can create pressure to respond quickly.

Note: A keyword is only a signal. Legitimate emails can contain these words too, so detection should consider context and multiple indicators.

16. Delivery and Parcel Spam

Delivery-themed phishing messages may claim that a package is delayed, held, or awaiting a fee. The recipient should verify tracking information through the courier's official website or app.

Important Keywords / Signals

No.	Keyword / Phrase	Why it may be suspicious
1	PACKAGE DELAYED	Attention-grabbing or promotional language.
2	DELIVERY FAILED	May create a reward-based incentive.
3	TRACK YOUR PACKAGE	Can create pressure to respond quickly.
4	CUSTOMS FEE	May request action, payment, or information.
5	DELIVERY FEE	Could be used in misleading offers or scams.
6	ADDRESS CONFIRMATION	Attention-grabbing or promotional language.
7	PARCEL HELD	May create a reward-based incentive.
8	RESCHEDULE DELIVERY	Can create pressure to respond quickly.

Note: A keyword is only a signal. Legitimate emails can contain these words too, so detection should consider context and multiple indicators.

17. Subject-Line Warning Patterns

Spam filters can examine unusual subject-line patterns. Excessive capitalization, repeated punctuation, suspicious promises, and combinations of urgency and money-related words can increase suspicion.

Important Keywords / Signals

No.	Keyword / Phrase	Why it may be suspicious
1	URGENT!!!	Attention-grabbing or promotional language.
2	FREE!!!	May create a reward-based incentive.
3	ACT NOW!!!	Can create pressure to respond quickly.
4	YOU ARE A WINNER	May request action, payment, or information.
5	FINAL WARNING	Could be used in misleading offers or scams.
6	IMPORTANT NOTICE	Attention-grabbing or promotional language.
7	CONGRATULATIONS!!!	May create a reward-based incentive.
8	CLAIM NOW	Can create pressure to respond quickly.

Note: A keyword is only a signal. Legitimate emails can contain these words too, so detection should consider context and multiple indicators.

18. Example Spam vs Legitimate Email

Example suspicious message: 'Congratulations! You are a winner. Claim your FREE prize now by clicking the link and confirming your account.'

Signals: prize claim, free offer, urgency, link request, account confirmation.

Example legitimate-style message: 'Your monthly statement is available. Sign in through the official app to review it.' The second message is not automatically safe, but it contains fewer obvious spam signals. Always verify unexpected messages.

Important Keywords / Signals

No.	Keyword / Phrase	Why it may be suspicious
1	winner	Attention-grabbing or promotional language.
2	free prize	May create a reward-based incentive.
3	claim now	Can create pressure to respond quickly.
4	confirm account	May request action, payment, or information.
5	click link	Could be used in misleading offers or scams.

Note: A keyword is only a signal. Legitimate emails can contain these words too, so detection should consider context and multiple indicators.

19. How Keyword-Based Detection Works

A basic keyword detector can tokenize the email, normalize text, compare words against a suspicious-keyword list, assign weights, and calculate a score. Example: 'free prize' could receive a higher score than the word 'free' alone.

Modern spam detection usually combines keyword signals with machine learning, sender reputation, URL reputation, message structure, authentication results, and user feedback.

Important Keywords / Signals

No.	Keyword / Phrase	Why it may be suspicious
1	keyword matching	Attention-grabbing or promotional language.
2	tokenization	May create a reward-based incentive.
3	spam score	Can create pressure to respond quickly.
4	weighted keywords	May request action, payment, or information.
5	machine learning	Could be used in misleading offers or scams.
6	sender reputation	Attention-grabbing or promotional language.
7	URL reputation	May create a reward-based incentive.

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20. Summary and 20-Key Keyword Checklist

Spam detection should never depend on one keyword. The strongest approach combines multiple signals and checks the context of the complete email. Users should avoid clicking unexpected links, opening suspicious attachments, or sharing passwords and verification codes.

A useful starter checklist contains: FREE, WINNER, PRIZE, URGENT, ACT NOW, CLAIM, BONUS, DISCOUNT, VERIFY ACCOUNT, PASSWORD, LOGIN, PAYMENT, REFUND, INVESTMENT, GUARANTEED PROFIT, WORK FROM HOME, CLICK HERE, DOWNLOAD NOW, INVOICE, and DELIVERY FAILED.

Important Keywords / Signals

No.	Keyword / Phrase	Why it may be suspicious
1	FREE	Attention-grabbing or promotional language.
2	WINNER	May create a reward-based incentive.
3	PRIZE	Can create pressure to respond quickly.
4	URGENT	May request action, payment, or information.
5	ACT NOW	Could be used in misleading offers or scams.
6	CLAIM	Attention-grabbing or promotional language.
7	BONUS	May create a reward-based incentive.
8	DISCOUNT	Can create pressure to respond quickly.
9	VERIFY ACCOUNT	May request action, payment, or information.
10	PASSWORD	Could be used in misleading offers or scams.
11	LOGIN	Attention-grabbing or promotional language.
12	PAYMENT	May create a reward-based incentive.
13	REFUND	Can create pressure to respond quickly.
14	INVESTMENT	May request action, payment, or information.
15	GUARANTEED PROFIT	Could be used in misleading offers or scams.
16	WORK FROM HOME	Attention-grabbing or promotional language.
17	CLICK HERE	May create a reward-based incentive.
18	DOWNLOAD NOW	Can create pressure to respond quickly.
19	INVOICE	May request action, payment, or information.
20	DELIVERY FAILED	Could be used in misleading offers or scams.

Note: A keyword is only a signal. Legitimate emails can contain these words too, so detection should consider context and multiple indicators.